

A certified hot-spots domain beyond the proven classes

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Abstract

Let Ω be the trapezoid with vertices $A = (0,0)$, $B = (1,0)$, $C = (17/20, 9/10)$, $D = (1/4, 9/10)$: convex, with side slopes 6 and $18/5$, no axis of symmetry, and not a lip domain. We prove, by a certified computation whose every step is either an exact rational decision or an outward-rounded interval enclosure, that the second Neumann eigenvalue of Ω is simple, with

μ_1 in $[12.020976137, 12.022398349]$,

and that the second Neumann eigenfunction attains its maximum and its minimum on the boundary only — and, more precisely, that the maximum is attained at vertex A and only there. To our knowledge this is the first certified hot-spots domain outside every class for which the conjecture was previously proven: all triangles (Judge and Mondal, Annals of Mathematics 2020, with the 2022 erratum), lip domains (Atar and Burdzy), certain non-convex L-tiled polygons (Hatcher, arXiv:2405.19508), and the symmetric quadrangle subcases (Deng-Gui-Jiang-Yang-Yao, arXiv:2604.19003). In the other direction, the conjecture is FALSE for convex sets in sufficiently high dimension (de Dios Pont, arXiv:2412.06344), which makes the planar convex case — where this domain lives — exactly the place the conjecture remains expected. The claim is one domain and one theorem; the convex-quadrilateral conjecture itself remains open. The certificate chain assumes two quoted results from the literature (Liu's lower-bound framework and the Crouzeix-Raviart interpolation constant $0.1893 h_K$, both from arXiv:1808.08148, whose PDF is pinned by sha256 in the repository) and standard facts; everything else re-derives mechanically in about two minutes.

1. The problem and its fences

Rauch's 1974 conjecture — the "hot spots" conjecture — asks whether the second Neumann eigenfunction of a bounded domain attains its extrema on the boundary: heat left to itself should push its hottest point to the edge. Burdzy and Werner refuted the general form with a domain with holes; for simply connected, and in particular convex, plane domains the conjecture is believed true and proved only classwise:

- **lip domains** (two Lipschitz-1 graphs): Atar-Burdzy, by probabilistic coupling;
- **all triangles**: Judge-Mondal, Annals of Mathematics 2020 (erratum 2022) — one Annals paper for the simplest polygon class, after partial acute-triangle results (Siudeja, arXiv:1308.3005);
- **certain non-convex polygons** (L-tiled domains): Hatcher, arXiv:2405.19508;

- **quadrangles with a symmetry axis** and related subcases: Deng-Gui-Jiang-Yang-Yao, arXiv:2604.19003 (April 2026).

And in the opposite direction: **the conjecture is false for convex sets in sufficiently high dimension** (de Dios Pont, arXiv:2412.06344, "Convex sets can have interior hot spots"), so the planar convex case is precisely where the conjecture remains expected. Convex quadrilaterals in the plane remain open. A validated-numeric route to acute triangles was developed in the Polymath7 project (with numerics by Nigam) before the analytic triangle proof; we cite it as the computational antecedent of the present approach. The specimen here is chosen to sit outside every listed class: its side slopes (6 and 18/5) exceed 1, so it is not a lip domain in any axis orientation; it has no symmetry axis; it is a quadrilateral, not a triangle; it lives in the plane. The interior angles are 74.48 deg (A), 80.54 deg (B), 99.46 deg (C), 105.52 deg (D); the corner exponents $\nu = \pi/\omega$ are 2.4169, 2.2350, 1.8097, 1.7058.

A certified computation is a different kind of object from a numerical study: every displayed inequality is a theorem about the exact eigenfunction, obtained through outward-rounded interval arithmetic over IEEE doubles, exact BigInt rational decisions where the statement is algebraic or geometric, and directed dyadic big-float enclosures for the independent cross-checks. Floating point proposes; it never decides.

2. The theorem

Theorem. Let Ω be the trapezoid above and let ϕ be a second Neumann eigenfunction of $-\Delta$ on Ω (μ_1 is simple, so ϕ is unique up to scale). Then ϕ attains its maximum and its minimum on the boundary of Ω only, and

$$\begin{aligned} \mu_1 &\text{ in } [12.020976137, 12.022398349] \quad (\text{width } 1.42\text{e-}3), \\ \mu_2 &\geq 13.955936. \end{aligned}$$

Normalization: $\hat{\phi} := c_1 \phi_1 = u - e$ as produced by the chain; extrema locations are scale-invariant; the sign is fixed by $\hat{\phi}(A) > 0$.

Corollary (the hot spot is vertex A). The maximum of $\hat{\phi}$ over the closure is attained at vertex A and only there, with

$$\hat{\phi}(A) \text{ in } [2.126029, 2.193158] \quad (= b_0(A) \text{ exactly}).$$

Proof from the chain's records: the interior witness w_+ lies inside A's corner sector (an exact rational disk decision); the certified radial monotonicity $d_r \hat{\phi} < 0$ on the whole punctured sector means $\hat{\phi}$ strictly decreases along every ray from A, so $\hat{\phi}(A) > \hat{\phi}(w_+) \geq 2.126029$; and every point outside the sector — core cells, collar cells, and the value ranges of tips B, C, D — is certified strictly below that bound. At the vertex the corner expansion collapses to $b_0(A)$ since $J_{\{k \nu\}}(0) = 0$ for $k \geq 1$. For triangles, "extrema only at vertices" is Judge-Mondal's refinement of the hot-spots statement; the maximum-side analogue now holds, certified, for this quadrilateral.

The minimum's location is deliberately left as an open question of enclosure width: $\hat{\phi}(C) = b_0(C)$ in $[-2.0204, -1.9789]$ overlaps the observed boundary minimum (float value -1.9998 at

distance 0.0195 from C along the top edge), so vertex-vs-edge-interior is not decided. A tighter corner extraction would decide it; an off-vertex answer would contrast with the triangle behaviour.

3. The certificate chain

The proof is a chain of eight machine-checked records; each stage reads its inputs from the upstream record, so no constant travels by hand.

3.1 Spectrum (certs/ember-spectrum.json)

Two-sided localization. Upper bounds by an interval Galerkin Rayleigh-Ritz method on the pulled-back unit square (basis $\cos(i\pi u) P_{\sim j}(v)$, shifted Legendre, $NU = NV = 12$): the v -integrals are exact BigInt rationals (the only non-polynomial piece is expanded in a positive series with a certified tail; intervals enter LAST), the u -integrals are closed forms over an interval π , and any float-chosen trial subspace gives a certified upper bound through a Gershgorin pencil estimate. Lower bounds by the Crouzeix-Raviart method: exact-rational assembly on the mapped grid (456 CR degrees of freedom; the kernel $K \equiv 0$ is checked exactly), discrete eigenvalue counts by interval LDL^T inertia with diagonal pivoting (Sylvester), and Liu's framework

$$\lambda_k \geq \lambda_{h,k} / (1 + C_h^2 \lambda_{h,k}),$$

$$C_h = 0.1893 h_{\max} \quad [\text{the two literature inputs}],$$

giving

$$\mu_1 \in [11.892662, 12.041820], \quad \mu_2 \geq 13.955936.$$

Since $13.9559 > 12.0418$, exactly one nonzero eigenvalue lies below 13.9559: **μ_1 is simple**. Calibration: the same code encloses μ_1 of the $1 \times 9/10$ rectangle around the exact π^2 at every run.

3.2 The trial and its boundary defect (certs/ember-defect.json)

A Method-of-Particular-Solutions trial $u = \sum a_i \psi_i$ over four corner Fourier-Bessel fans, $\psi_i = J_{\{k, \nu\}}(\sqrt{\lambda_{\sim}} r) \cos(k \nu \theta)$, with $\lambda_{\sim} = 12.021687243$ and 40 frozen exact-double coefficients from a deterministic float proposer (Betcke-Trefethen subspace angle, rank-truncated SVD, an absolute noise floor for J_{ν} at large ν). The trial satisfies $-\Delta u = \lambda_{\sim} u$ EXACTLY, and each fan's conormal derivative vanishes analytically on its own corner's two edges, so only far-corner fans produce boundary flux. Along each edge the flux closes under differentiation via the Bessel ODE, so order-2 interval Taylor jets give (f, f', f'') with a single hand formula, and a certified midpoint-Taylor cell rule over 700 cells per edge yields

$$\|d_{\nu} u\|_{L^2(\text{boundary})} \leq 1.2369e-5$$

(float value $9.01e-6$ — the certified overhead is modest), plus certified per-edge pointwise sups $|d_{\nu} u| \leq 1.39e-5$. Twelve value bridges and four derivative bridges against an independent float evaluation guard the jets (a passing value bridge does not test a jet).

3.3 The eigenpair (certs/ember-eigenpair.json)

For the exact-Helmholtz trial, Green's identity gives $r(v) = a(u,v) - \lambda \sim m(u,v) = \text{int_boundary} (d_{\nu} u) v \, ds$, so with the rational star-shaped trace constant (star center $x_0 = (21/40, 9/20)$, $c_0 \geq 0.38541$, $R \leq 0.69147$, $C_{\text{tr}} \leq 2.64262$):

$$\sum_k c_k^2 (\mu_k - \lambda \sim)^2 / (1 + \mu_k) \leq \epsilon^2, \\ \epsilon = D \, C_{\text{tr}} = 3.2686e-5.$$

With $F = \max(1/\lambda \sim^2, (1+\mu_2)/(\mu_2-\lambda \sim)^2) = 3.9975$ and a certified interior-box lower bound $\|u\| \geq 0.166$:

$$\mu_1 \text{ in } [12.020976137, 12.022398349], \\ \|u - c_1 \phi_1\|_{L^2} \leq 6.5352e-5 \quad (\text{relative } 3.94e-4), \\ |e|_{H^1} \leq 2.4414e-4, \quad |\lambda \sim - \mu_1| \leq 7.1111e-4.$$

This tightens the CR window by a factor of about 105 and is the Moler-Payne mechanism in its modern a-posteriori form.

3.4 The partition, decided in rationals

The interior is partitioned on the $1/100$ grid: a cell is a CORE cell iff the minimum of the boundary-distance over its four corners is $\geq 3/40$ — exact, because the distance-to-boundary of a convex domain is concave, so its minimum over a cell is attained at a corner; a cell belongs to a corner sector iff it lies ENTIRELY inside that vertex's 0.11 -disk — exact, because $|p - V|$ is convex, so its maximum over a cell is attained at a corner; the remaining cells meeting Ω (an exact separating-axis decision) are COLLAR cells. The census — 4690 core, 2272 collar, 498 sector cells — is re-decided in exact rationals at assembly time. Every interior point within 0.11 of a vertex lies in that vertex's sector by convexity, so the three classes cover the interior by construction.

3.5 Core and collar sweeps (certs/ember-pointwise.json, certs/ember-collar.json)

The pointwise machinery is a solid-mean lemma with the EXACT kernel norm: for $-\Delta w = f$ on $B_R(x_0)$,

$$w(x_0) = \text{avg}_{B_R} w + \text{int}_{B_R} G_R f, \quad \|G_R\|_{L^2} = R \sqrt{I_0 / 2 \pi}, \\ I_0 = \text{int}_{0^1} (-\ln t - (1-t^2)/2)^2 t \, dt = 5/48 \quad (\text{derived in exact rationals}).$$

Two interior witnesses anchor the argument, their balls decided inside Ω in exact rationals:

$$\phi\text{-hat}(w+) \geq 2.126029 \quad (\text{near } A, \text{ ball } R = 0.0318), \\ -\phi\text{-hat}(w-) \geq 1.993811 \quad (\text{under the boundary min, } R = 0.018).$$

Every CORE cell is killed on both sides by whole-cell value bounds plus the solid-mean error at $R = 0.07$ (valid at every core point), with kill-or-refine to $1/400$: zero survivors, margins $3.93e-2$ (max side) and $5.61e-4$ (min side). Every COLLAR cell is killed the same way with REFLECTED bounds: the even reflection of $\phi\text{-hat}$ across the nearest open edge is exact, and the reflection of u adds a single layer with density $2 \, d_{\nu} u$, bounded by $\sup |d_{\nu} u| \, 3R/\pi$ from the per-edge certified sups — 2272 cells, zero survivors (worst kills 2.123592 vs 2.126029 and 1.993633 vs 1.993811).

3.6 The corner sectors (certs/ember-corner.json)

In each corner sector, Neumann separation of variables and H^1 regularity (which excludes the singular Bessel family) give EXACTLY

$$\hat{\phi} = \sum_k b_k J_{\nu}(k r) \cos(k \theta).$$

The coefficients b_0, b_1, b_2 are certified by big-annulus L^2 extraction — the corner's own fan integrates exactly by angular orthogonality, the far part by second-order midpoint cells, the eigenfunction error contributes at most $E \sqrt{D_k}$ — with a Parseval bound on everything above mode 2. **The extraction is the chain's most delicate step, so it is performed at TWO annuli, and the two enclosures of every coefficient must intersect (they do; this is a condition of entry for the theorem record):**

corner	ν	b_0 (annulus 1)	b_0 (annulus 2)	b_1 (annulus 1)	b_1 (annulus 2)
A	2.4169	[2.109, 2.194]	[2.121, 2.182]	[2.682, 2.988]	[2.691, 2.978]
B	2.2350	[0.831, 0.878]	[0.837, 0.871]	[-4.596, -4.273]	[-4.575, -4.293]
C	1.8097	[-2.021, -1.978]	[-2.016, -1.984]	[-2.046, -1.893]	[-2.036, -1.903]
D	1.7058	[-1.571, -1.546]	[-1.568, -1.549]	[3.418, 3.634]	[3.433, 3.618]

On each sector $r \leq 0.11$ (the wedge at C to 0.12):

- **B and D die by value on both sides:** $\hat{\phi}$ ranges [0.756, 0.897] and [-1.665, -1.351] — clear of both witnesses.
- **A** (the boundary maximum lives at the vertex): the min side dies by value (range [2.015, 2.194]); on the max side an interior maximum would force a critical point, but $d_r \hat{\phi} < 0$ throughout the sector (worst $-4.99e-2$ on r in [0.008, 0.11]; on $r \leq 0.008$ the factored form $d_r \hat{\phi} / r \leq -10.04 < 0$).
- **C** (the boundary minimum lives 0.0195 from the vertex): the max side dies by value (range upper -1.847). The min side splits at $\theta_1 = 0.35$. On [0.35, ω], the gradient min-form — a quadratic in $\cos(\nu \theta)$ with mode 2 and the tail as perturbations — stays positive down to $r = 1e-5$ (worst 0.0001), and an analytic limit piece covers $r \leq 1e-5$. On the wedge [0, 0.35] along the top edge, the second tangential derivative comes from the Bessel ladder identity

$$d_t^2 [J_\nu(k r) \cos(\nu \theta)] = (k^2/4) [J_{\nu+2} \cos((\nu+2) \theta) + J_{\nu-2} \cos((\nu-2) \theta) - 2 J_\nu \cos(\nu \theta)],$$

which is exact and sign-explicit where the polar-split pieces diverge individually: the singular term $J_{\nu-2}$ (a NEGATIVE fractional order, $\nu - 2 = -0.190$) arrives multiplied by the certified $b_1 < 0$, so it HELPS. The result is $\hat{\phi}_{nn} \geq 13.04$ across the whole wedge down to $r = 1e-6$ (inner piece ≥ 11.65), and the exact second-order Taylor expansion from the Neumann edge (the first-order term vanishes by the boundary condition) forces every interior wedge point strictly above the boundary minimum.

3.7 Assembly

Suppose an interior x attains the maximum of $\hat{\phi}$. Then $\hat{\phi}(x) \geq \hat{\phi}(w^+) \geq 2.126029$. But x lies in a killed core cell, a killed collar cell, or a corner sector: B, C, D are dead by value, and in A

a maximum forces grad phi-hat(x) = 0, contradicting $d_r \phi\text{-hat} < 0$. So the maximum lives on the boundary only; the minimum argument is symmetric with $w\text{-'}$. QED.

3.8 Independent cross-derivations (certs/ember-cross.json)

Two implementations, one gate: $I_0 = 5/48$ re-derived in exact rationals; the trace constant re-derived from the exact-rational star geometry on directed dyadic big-floats, C_{tr} in $[2.642618944993, 2.642618944993]$ (the doubles route agrees); μ_1 bounded above independently on a conforming P1 finite-element basis (12.0642 at $n = 16$ — a different discretization family confirming the localization from above); and the second-annulus corner extraction of section 3.6.

4. Methods notes (the lessons the falsifiers taught)

1. Intervals enter LAST in rational-core pipelines: converting per-term against large Legendre coefficients destroys exact cancellation. 2. Unpivoted interval LDL^T dies mid-elimination; diagonal-pivoted right-looking elimination with float pre-scaling (both congruences) is robust. 3. Galerkin strong residuals diverge at corners; exact Helmholtz corner fans make the interior residual identically zero and leave only boundary flux — the architecture the corner exponents force. 4. A passing VALUE bridge does not test a jet: bridge every derivative order used (a $\theta\text{'-sign}$ forgery inflates the defect a thousandfold while all value bridges pass — the battery keeps this as a red). 5. Cell tests must match the geometry: depth is concave (cell minimum at a corner — exact), distance is convex (cell maximum at a corner — exact); a partition must be re-decided exactly, never assumed from sweep bookkeeping. Two quantization slivers in an earlier version of the partition logic were found and closed this way; the battery keeps the unsound rule as a firing red control. 6. The ladder identity beats the polar split wherever individually divergent pieces cancel: choose the form whose signs the interval arithmetic can see.

5. Lineage

The Method of Particular Solutions is Fox-Henrici-Moler (1967); the eigenfunction enclosure from a boundary defect plus a spectral gap is Moler-Payne (1968), used here in its residual form with the certified Crouzeix-Raviart gap as the gap provider (Liu's framework, You-Xie-Liu arXiv:1808.08148); the subspace-angle regularization of MPS is Betcke-Trefethen (2005). The corner analysis is the computational twin of Judge-Mondal's. [OPERATOR/CITE: verify the final reference list against the pinned PDFs before submission.]

6. Reproducibility and trust base

Everything re-runs from the repository, deterministically:

```
node tools/run-ember-chain.js      # all 8 stages, ~2 min
node instruments/hotspots/battery.js # record walk + 8 red controls
node instruments/ivspecial/battery.js # the Gamma/Bessel layer
make test                          # every battery in the machine
```

Trust base, stated so nothing hides: the interval rigor model (outward-rounded IEEE doubles; exact BigInt rationals; directed dyadic big-floats for cross-checks); the interval special-function layer with its closed-form falsifiers and two-implementation gates; TWO literature inputs — Liu's framework

theorem (Thm 2.4) and the Crouzeix-Raviart constant $0.1893 h_K$ (Lemma 3.2), both quoted from arXiv:1808.08148, PDF pinned sha256 fb867aa5ab937b92..., transcription beside the pin — and classical facts (sector Neumann separation + H^1 regularity, Green's identity, the spectral theorem, Courant). Machine-derived; not peer-reviewed; not independently rerun.

7. Open problems

1. **The census.** Sweep the moduli space of convex quadrilaterals with interval-coefficient shape boxes toward "convex quadrilaterals have no hot spots". Every instrument above is parameterized by the quadrilateral; the campaign is scoped, priced, and NOT started — and is not counted here. 2. **The cold spot's exact location.** Decide vertex C versus the edge interior (see the corollary's open twin): a tighter $b_0(C)$ extraction plus one certified edge comparison settles it either way. 3. **Sharper μ_1 .** The enclosure width $1.42e-3$ is defect-limited; larger fans and finer edge quadrature buy digits directly. 4. **Other single domains.** Non-trapezoidal convex quadrilaterals with all slopes > 1 need no new mathematics, only compute.

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